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80/100

CO1: Assessment Tool-1

1) Event communication Security

a) Encrypt the message "MEET AT NOON" using caesar cipher with a shift of 4.

The caesar cipher is one of the earliest and simplest classical encryption techniques. It is a substitution cipher in which each letter of the plaintext is replaced by another letter that is fixed number of positions ahead in the alphabet. In this scenario, the shift value is 4, means every letter moves four places forward while spaces remain unchanged.

Encryption Process-

Plain text = M E E T A T N O O N

Shift (+4) = Q I I X E X R S S R

Cipher text - QIIX EX RSSR

The original message is securely transformed into an unreadable form using a shift of four positions.

b) Retrieval of the original message -

An unauthorized receiver decrypts the cipher text by shifting every letter 4 positions backw

Example - Q → M

I → E

X → T

Applying this process to complete cipher text:

QIIX EX RSSR → MEET AT NOON

Only user who know the shift value can correctl recover the original message.

2) Intercepted suspicious message

Cipher text : GSRH RH Z HVXIVG

a) Original message

The message is encrypted using the substitution cipher, where each alphabet is replaced by its opposition alphabet.

Example -

G → T

S → H

R → I

H → S

Decoded message : THIS IS A SECRET

b) Justification

The message follows a simple substitution pattern rather than a complex encryption algorithm because:

-> Repeated words remain repeated after substitution.

-> 'RH' appears twice and decodes to 'IS'.

-> 'Z' becomes the single-letter word 'A'.

-> The last word decodes to 'SECRET', which is meaningful english.

The cipher text follows the substitution technique and the original message is "THIS IS A SECRET".

3) Secure messaging system.

Plaintext = HELLO

Keyword = KEY

a) Encryption Process

Repeat the keyword until it matches the message length.

Plaintext - H E L L O

Keyword - K E Y K E

Assign values (A=0 to Z=25)

Plain -	H	E	L	L	O
value	7	4	11	11	14

Key =	K	E	Y	K	E
Value	10	4	24	10	4

Apply, cipher = (Plain + Key) mod 26

Calculation	Result
7+10 = 17	R
4+4 = 8	I
11+24 = 35 → 9	J
11+10 = 21	V
14+4 = 18	S

Ciphertext : RIJVS

b) Effect of the keyword

The keyword determines the shift value for every character.

1) Different Keyword letters produce different shifts.
2) The Keyword repeats until the message ends.
3) Even identical plaintext letters may encrypt differently if different keyword letters are used.

The repeating keyword increases security compared to the caesar cipher because multiple shift values are used.

4) Suspicious Digital Asset Investigation.

a) Detecting Hidden Data

Investigators can detect hidden information by:

1) Performing steganalysis.
2) Examining image metadata.
3) Comparing the suspected image with the original image.
4) Analyzing Least significant Bits (LSB)
5) Using forensic tools such as stegexpose/stegsolve.

These methods help determine whether confidential data embedded inside the image.

b) Strengthening Security

Two techniques are:

1) Encrypt the message before embedding.

Even if extracted, the message cannot be read without the encryption key.

2) Use advanced steganography

Hide data using randomized pixel locations with password protection, making extraction difficult.

Combining encryption with steganography provides stronger confidentiality.

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5) Legacy Encryption in Corporate Communication.

Cipher text - WECRLTEERDSOEEFEAOCAIVDEN

a) Original message

The ciphertext is produced using a 3-Rail Fence cipher.

After reversing the transposition,

Original message -

WE ARE DISCOVERED FLEE AT ONCE

b) Why is this method vulnerable?

1) It uses only rearrangement of letters without changing them.

2) Letter frequencies remain unchanged.

3) It is vulnerable to brute-force attacks.

4) modern computers can break it quickly

Rail Fence cipher is not suitable for protecting confidential corporate communication.

c) Practical Enhancement

Replace the Rail Fence cipher with a modern encryption algorithm such as AES-256.

Advantages:

→ strong key-based encryption.

→ Resistant to brute-force

→ widely used in banking, government and secure communication systems.

Modern encryption algorithms like AES provide significantly better confidentiality and security than classical transposition ciphers.